# LINUS LIU

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## EDUCATION

**Rice University** Houston, TX | Expected May 2028
B.S. Computer Science; B.A. Computational and Applied Mathematics
**Mathematics Major GPA: 4.00/4.00** | **Overall GPA: 3.32/4.00**
**Coursework:** Algorithms; Machine Learning; Honors Linear Algebra and Multivariable Calculus.

## TECHNICAL SKILLS

**Programming:** Python, Java, TypeScript, SQL; C++20 (current research project)
**AI & data tools:** PyTorch, JAX, Gemini API, pandas, NumPy, scikit-learn, LightGBM, HDF5
**Engineering:** API integration, caching, hybrid retrieval, differential testing, greedy algorithms, React, Vite

## AI & ENGINEERING EXPERIENCE

**WISHEE** Dec 2025 – Present
*Intern | Voice Assistant Retrieval | Shenzhen / Remote*

- Built a cloud retrieval layer over **30,000 documents and 600K chunks**, serving **20 queries/sec** at **600 ms p95 first-token latency** for a smart-earbud voice assistant.
- Cached speculative searches on partial speech transcripts, cutting post-speech wait from **680 ms to 440 ms**.
- Combined dense vectors, **BM25**, metadata filters, and reciprocal rank fusion, improving **recall@5 from 0.68 to 0.89** on **200 hand-labeled voice queries**.

**LENOVO** Apr – Aug 2025
*Machine Learning Intern | Sequential Decision Modeling | Beijing, China*

- Implemented a **Decision Pre-trained Transformer** for multi-zone HVAC control on IoT data; built reproducible **Python/JAX training and offline evaluation workflows**.
- Ran **context-length and regularization ablations**; used **SQL filtering**, metric analysis, and replay-based testing to investigate failure cases and model stability.

**MD ANDERSON CANCER CENTER** Mar – Sept 2026
*Researcher | Computational Pathology | Advisor: Andrew Song*

- Co-developed a **whole-slide image pipeline**: tissue segmentation, patch extraction, **CONCH/Virchow2** encoding, and **CLAM-compatible HDF5** feature bags for downstream modeling.
- Developing a **JEPA-style world model** to predict future pathology embeddings from actions and elapsed time.

## SELECTED SOFTWARE PROJECTS

**HACKRICE – GAME2LEARN** Sep 2025 – Present
*Lead Full Stack Developer | Learning Application | GitHub*

- Led a **five-person team** building a **React/TypeScript** app; co-developed **Gemini-powered PDF/video analysis** and scene-based quizzes with answer feedback.
- Co-developed a **two-stage Gemini API workflow** from video to application specification to HTML, with a **Monaco editor**, **sandboxed iframe** preview, and **localStorage** save/reload.

**RICE QUANT FUND** Apr 2026 – Present
*Quant Researcher | Order-Book Collector & Replay Study | In Progress*

- Developing a **C++20** market-data collector and order-book engine with fixed-point prices and quantities, receive-time sampling, and a **bounded producer-consumer queue**.
- Designed snapshot recovery and positioned drop markers; specified **Python-reference differential tests** and synthetic archives with injected gaps, duplicates, and clock jumps.

**FEAT – AUTOMATED TEST GENERATION FRAMEWORK** Aug – Dec 2025
*Software Engineering Project | Rice University*

- Built a **Java differential-testing tool** to compare program behavior; formulated test-suite reduction as **set cover** and implemented **greedy test selection** targeting bug-detection coverage.

## SELECTED HONORS

**HiMCM** – Meritorious Award (2022); **S.-T. Yau High School Science Award** – Third Prize (2023).